UMER PATEL

Scientific Computing & AI-Native Software Engineer

India (Remote) | umerpatel1540@gmail.com | umerpatel.vercel.app | linkedin.com/in/umerpatel | github.com/callmeumair

PROFESSIONAL SUMMARY

- Scientific computing expert and AI-native iOS/full-stack engineer with a shipped App Store application
- Build and validate terminal-based scientific computing tasks in materials science for Turing, and evaluate frontier AI models and coding agents for Handshake AI and Mercor
- Strong in Python, Swift/SwiftUI, TypeScript, and SQL, with hands-on experience in RLHF evaluation, model output grading, and rubric-based quality review
- Daily user of Claude Code, Cursor, Windsurf, GitHub Copilot, and OpenAI Codex

SKILLS

- **Scientific Computing:** Python, NumPy, numerical modeling, data reduction, simulation workflows, reproducible computational tasks, Linux terminal and Docker environments
- **Languages:** Python, Swift, SwiftUI, TypeScript, JavaScript, SQL
- **AI and Evaluation:** LLM evaluation, RLHF data labeling, rubric-based grading, red-teaming, prompt engineering, LangChain, AI coding agents
- **iOS Development:** SwiftUI, MVVM, Mapbox, Firebase, Push Notifications, REST APIs, App Store deployment
- **Web and Backend:** Next.js, React, Node.js, Express, FastAPI, Tailwind CSS, Prisma, PostgreSQL, MongoDB
- **Tools and Testing:** Git, GitHub, CI/CD, Docker, Vercel, Postman, Jest, Playwright

PROFESSIONAL EXPERIENCE

Scientific Computing Expert (Contract, Remote) | Turing | Oct 2026 – Present

- Design and build terminal-based scientific computing tasks in materials science for the Terminal Bench Science (TB-Science) benchmark
- Implement numerical and data-analysis workflows in Python, including data reduction, physical modeling, and inversion problems, with deterministic and verifiable outputs
- Write automated verifiers, reference solutions, and task specifications that meet strict correctness and reproducibility standards
- Validate tasks against frontier AI models to confirm difficulty, then revise based on reviewer feedback until accepted on the evaluation platform
- Collaborate with project managers and pod leads to meet quality gates across multiple concurrent task batches

AI Model and Coding Agent Evaluator (Contract, Remote) | Handshake AI | Jul 2026 – Present

- Evaluate and rate frontier LLM outputs on software engineering tasks in Swift/SwiftUI, Python, TypeScript, and full-stack code generation, following detailed rubrics
- Perform side-by-side model comparisons, correctness verification, and quality scoring to support RLHF and SFT data pipelines
- Write and review edge-case test scenarios and identify hallucinations, logic errors, and security or performance issues in AI-generated code
- Maintain inter-rater consistency and meet throughput and quality SLAs across concurrent evaluation projects

AI Model Evaluation Contributor (Remote) | Mercor / AI Research Lab | 2024 – Present

- Evaluated frontier AI model outputs on coding tasks across multiple languages for correctness, code quality, maintainability, and runtime performance
- Contributed structured feedback to RLHF and model assessment pipelines used to fine-tune production LLMs
- Benchmarked agentic coding tools (Claude Code, Cursor, Windsurf, Copilot, Codex) on real iOS and full-stack tasks to surface systematic model weaknesses

PROJECTS

CommuteTimely – Smart Commute Planner (iOS) | Sep 2024 – Present

- Built and shipped an AI-powered iOS app to the App Store that predicts bus delays from historical data and real-time signals and recommends routes with live ETA
- Stack: SwiftUI, MVVM, Mapbox API, Python, FastAPI, Firebase, Push Notifications
- Used AI coding agents to scaffold MVVM architecture and SwiftUI components, then reviewed code for correctness, lifecycle safety, performance, and edge cases
- Implemented secure authentication, push notifications, and analytics

SwiftAgent Evaluator – AI Coding Agent Benchmarking Platform | May 2025 – Present

- Built a full-stack platform to benchmark Claude Code, Cursor, Windsurf, Copilot, and Codex on real-world iOS engineering tasks
- Implemented a Model Arena, live leaderboard, 3D score visualization, evaluation replay timeline, and animated code-diff viewer
- Stack: Next.js, TypeScript, Tailwind CSS, Python, Prisma, PostgreSQL, Three.js, deployed on Vercel with CI/CD

SecureSight Dashboard – Security Monitoring Platform | Jan 2025 – Present

- Built a full-stack dashboard for monitoring vulnerabilities and system health with real-time updates and visualizations
- Implemented secure authentication, role-based access control, and performance-optimized APIs
- Stack: Next.js, TypeScript, Node.js, Express, MongoDB, Chart.js, deployed on Vercel with CI/CD

PulseFit AI – AI Fitness Coaching App | Apr 2025 – Present

- Built an AI platform that generates personalized workout and diet plans using the OpenAI API
- Added an analytics dashboard tracking progress, streaks, and performance trends
- Stack: Next.js, Python, OpenAI API, Tailwind CSS, MongoDB, Vercel

EDUCATION

SRM Institute of Science and Technology | Chennai, India | 2024 – 2026

- Bachelor of Computer Applications (BCA), Data Science Major
- GPA: 9.0 / 10.0

ADDITIONAL

- Open Source Contributor (2024 – Present): contribute to open-source projects and collaborate with developers globally
- Tech Community (2024 – Present): active in hackathons and developer networks